

6th Grade Team Exam

2026 James Clemens Math Tournament

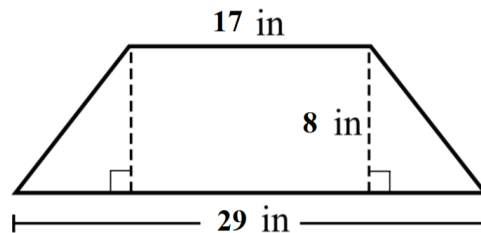
INSTRUCTIONS:

1. Your team has 60 minutes to complete this examination.
2. One official team from each school consisting of four students and other unofficial teams may compete. Only the scores from the official team will be counted towards a school's final score.
3. This exam consists of 15 free-response questions, each worth 10 points if answered correctly and 0 points if answered incorrectly or left unanswered.
4. Calculators, cellular devices, books, notes, and other aids are prohibited during this exam. Scratch paper will be provided for calculations. If you require additional scratch paper, simply raise your hand and your proctor will assist you.
5. Write your answers to the questions in the provided team answer sheet. You may use the test booklets for scratch work, but only the answers marked in the team answer sheet will be counted. Please write **legibly**.
6. **All answers to the team exam will be a positive integer from 1 to 999, inclusive.**
7. **Leave out** all units and special characters when recording your answer on the team answer sheet. For example, if you get \$1 as your answer, write 1 on the answer sheet.
8. A school's final score for a grade will be calculated by adding the top 4 written test scores in that grade (excluding tiebreaker questions) as well as the team round score of the official team, for a maximum of 550 points.
9. In the event of a tie, the school with the highest team round score will be favored. If team round scores result in another tie, answers will be evaluated starting backwards from question 15 to 1 to determine a winner.
10. Diagrams are not necessarily drawn to scale.
11. Although this math tournament is intended to demonstrate your knowledge and skills in math, be sure to enjoy the experience and have fun!
12. You may keep the test booklet after the exam.

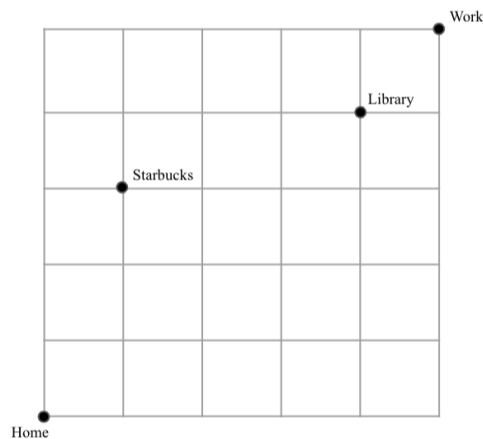
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1. Bob has 3 pairs of jeans, 5 pairs of socks, 7 t-shirts, and 2 pairs of shoes. How many different outfits can he make if each outfit consists of one pair of jeans, one pair of socks, one T-shirt, and one pair of shoes?
2. Anaya went to a haunted house and saw spiders and witches. If she counted 24 heads and 108 feet, how many spiders did she see?
3. Rakshan is driving to the grocery which is 10 miles away at a speed of 30 mph. He drives back home at a speed of 60 mph. In miles per hour, what is his average speed over the entire trip?
4. Isabella is buying a pair of shorts that was originally \$60, but is now on sale for 30% off. If Isabella also has to pay a 7% sales tax on the discounted price, rounded to the nearest dollar, how much money does she have to pay?
5. Colby is volunteering at his school by helping to sell culture night tickets. Each child ticket is worth \$5 and each adult ticket is worth \$7. If Colby sold \$530 in ticket sales and counted 86 total attendees, how many adults were at the event?
6. The number 102_3 is added to 10101_2 , and the result is written in base 10. What is the product of the digits of the resulting number?
7. Priscilla has a pile of pennies. First, she puts these pennies in groups of 6 and has 3 leftover. Next, she puts them in groups of 5 and has 1 penny leftover. Finally, she puts them in groups of 8 and has 5 leftover. What is the smallest number of pennies she could have?
8. Harriet, Jim, Lucy, Rachel, and Tom are all going to the movies. Rachel, Harriet, and Lucy have all gotten into an argument and do not want to sit next to each other (ex: Harriet cannot sit next to Rachel or Lucy). How many ways can the five of them be seated?

9. Let x be the number of positive factors of 360. Let y be the number of ways to distribute 8 distinguishable socks into 2 distinguishable bins, where each bin can have any number of socks. What is $x + y$?
10. Andy the ant is trying to fence his isosceles trapezoid-shaped yard. The two bases of his yard measure 17 in and 29 in and the distance between the two bases is 8 in, as shown in the diagram below. If one inch of fence costs \$2.50 to install, how much money, in dollars, will it cost him to install a fence around the entire perimeter of his yard?



11. How many ordered triples (a, b, c) satisfy the equation $abc = 36$ given that a, b , and c are all distinct positive integers?
12. Kassidy is trying to get to work. But, she wants to grab a coffee at Starbucks and pick up a book from the library on the way. If she can only move upward and rightward along the grid, how many paths can she take to get to work and complete all her stops?



13. A palindrome is a number that is the same when its digits are reversed. For example, 121 is a palindrome. How many palindromes are there between 100 and 1000?
14. How many more trailing zeros are at the end of $63!$ than there are at the end of $45!$?
15. In a standard deck of cards, Francis draws one card and then draws another without replacement. He calculates the probability that the first card is a queen and second card is a heart. If this probability can be expressed as a simplified common fraction in the form $\frac{a}{b}$, what is $b - a$?